



OSCILLATIONS, WAVES AND OPTICS

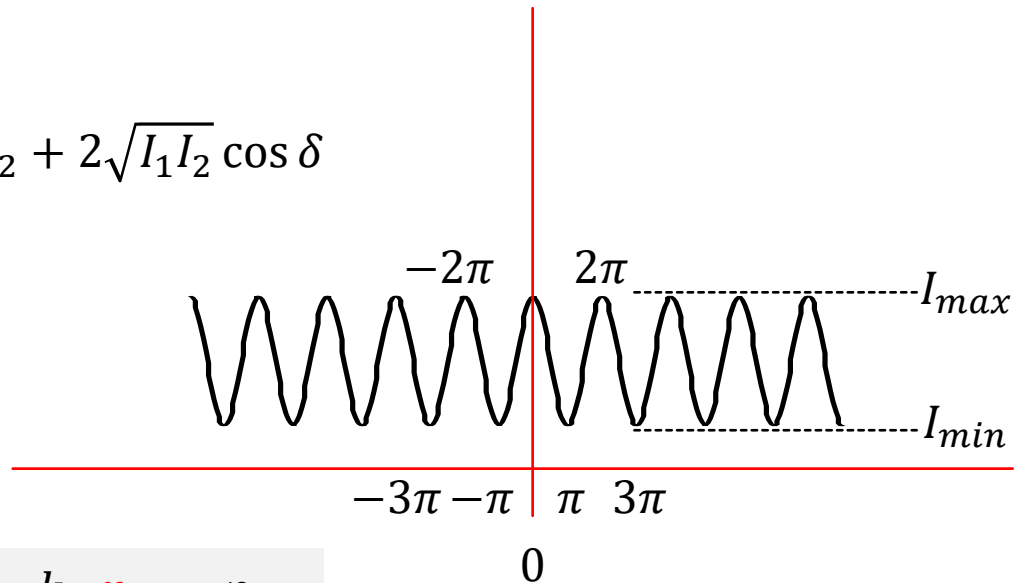
PHYSICS FOR CHEMISTRY STUDENTS

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CHAPTER 4. ELECTROMAGNETIC WAVES

REVIEW

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \delta$$



$$\delta = \alpha_{01} - \alpha_{02} = k \cdot \mathbf{r}_{01} + \varphi_{01} - k \cdot \mathbf{r}_{02} - \varphi_{02}$$

constructive $\delta = 2m\pi$

$$\Delta = m\lambda$$

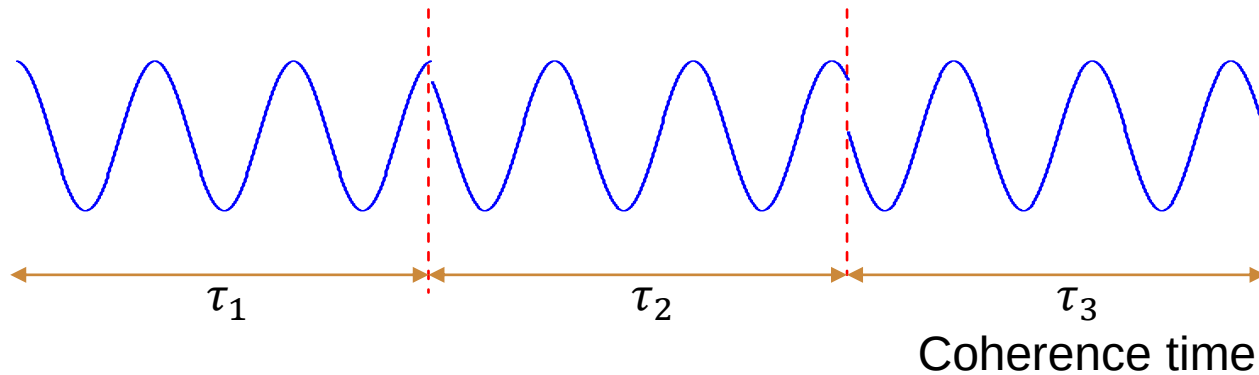
destructive $\delta = (2m + 1)\pi$

$$\Delta = \frac{2m + 1}{2} \lambda$$

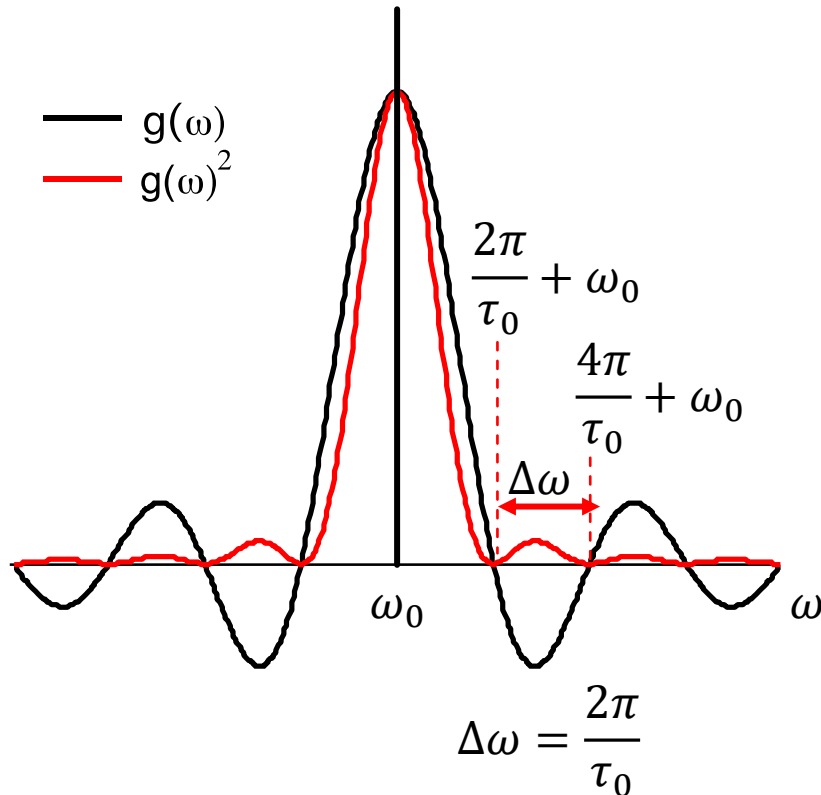
***Need to consider
phase shift***

REVIEW

“Monochromatic” light



REVIEW



$$g(\omega) = \frac{\tau_0}{2\pi} \frac{\sin u}{u} \quad u = \frac{(\omega - \omega_0)\tau_0}{2}$$

the shorter τ_0 , the wider $g(\omega)$

$$\tau_0 \rightarrow \infty, \omega \rightarrow \omega_0$$

QUIZ

$$\lambda = 600\text{nm}$$

$$E_1(r, t) = E_0 \sin(kz + \omega t + \varphi_0)_{\omega t + \alpha}$$

$$E_2(r, t) = E_0 \sin(kz + \omega t + \varphi_0)_{\omega t + \alpha}$$

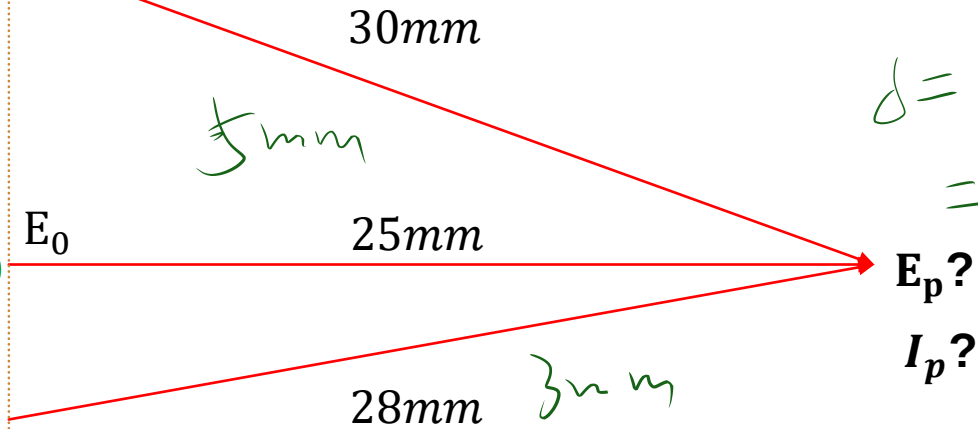
$$E_3(r, t) = E_0 \sin(kz + \omega t + \varphi_0)_{\omega t + \alpha}$$

$$E_p^2 = 3E_0^2 \sin^2 \omega t + 3E_0^2 \cos^2 \omega t$$

$$= 3E_0^2$$

$$I_p = \epsilon_0 c \langle E_p^2 \rangle = I_1 + I_2 + I_3 + 2I_2 + 2I_3$$

$$3I_0 + 2I_0(\cos \delta_1 + \cos \delta_2)$$



$$\delta = k \Delta z$$

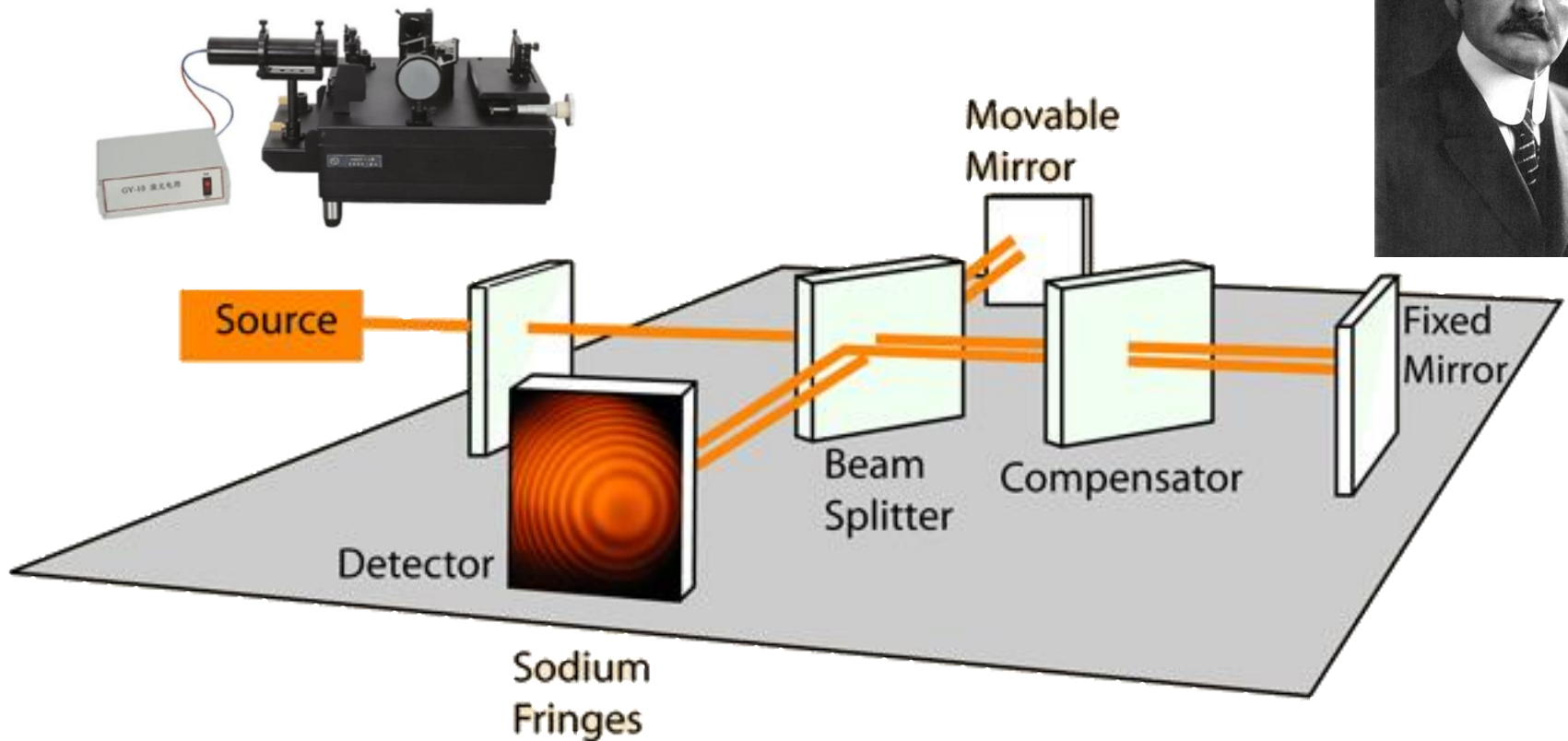
$$= \frac{2\pi}{600}$$

2mm

$$\delta_1 = \frac{5}{3} \pi \times 10^{-6}$$

4.5 INTERFEROMETERS

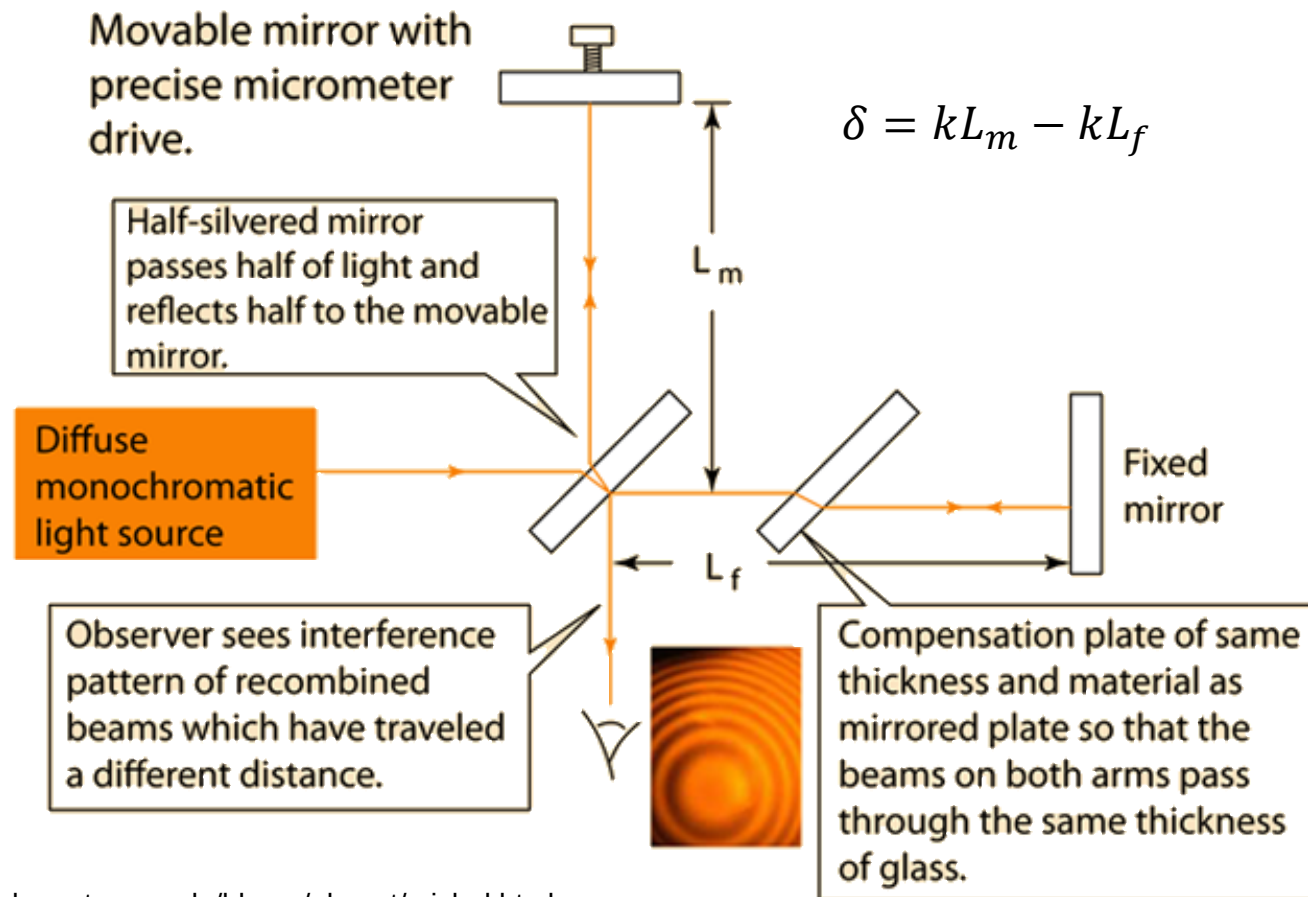
Michelson interferometer



4.5 INTERFEROMETERS

<https://www.youtube.com/watch?v=j-u3IEgcTiQ>

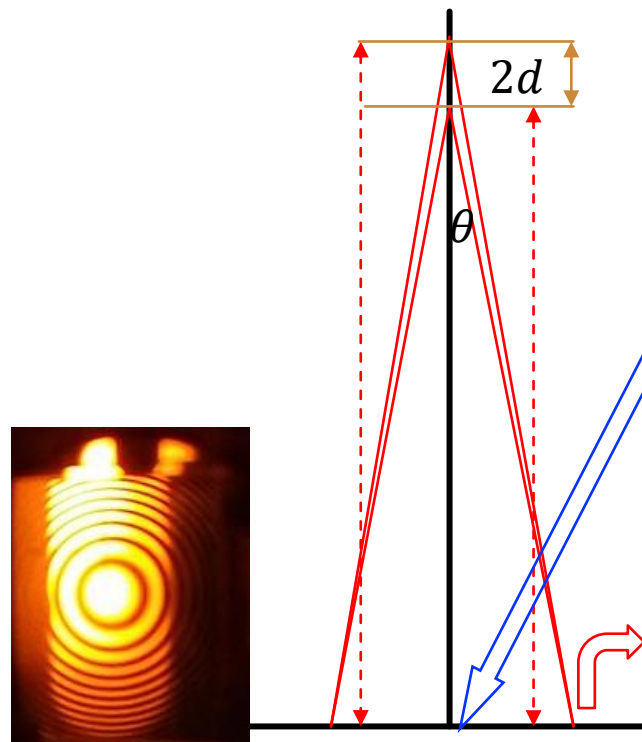
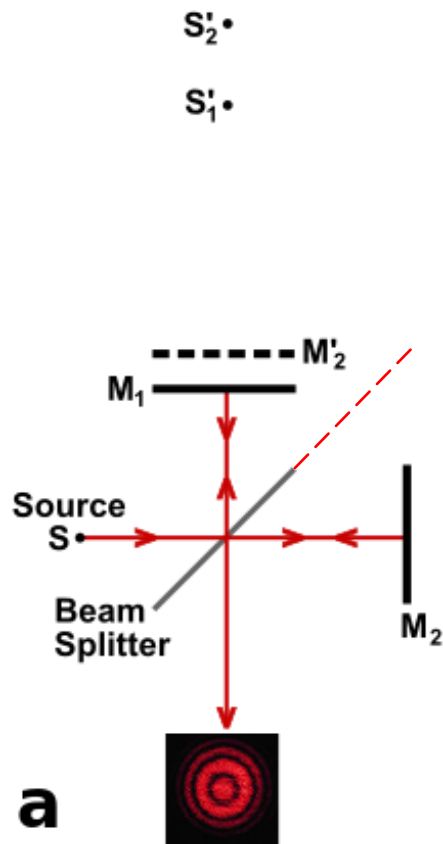
Michelson interferometer



4.5 INTERFEROMETERS

Michelson interferometer

Equivalent optics



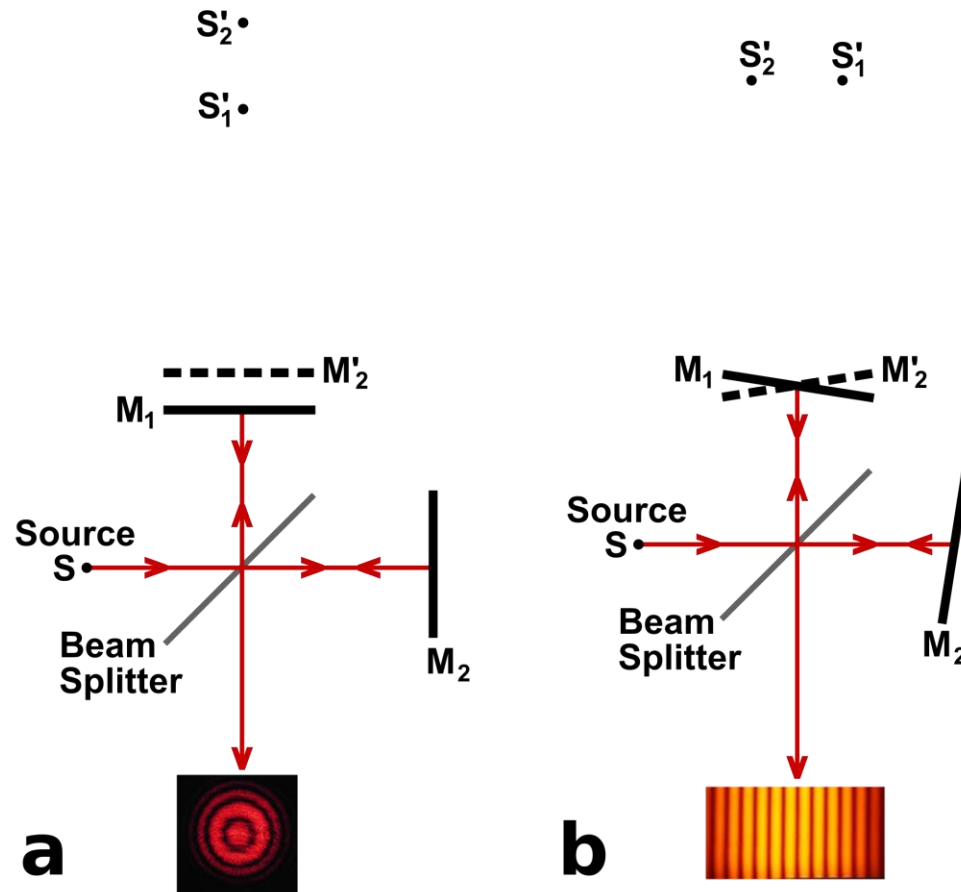
$$\delta = k2d$$

$$2d = m\lambda$$

$$2\Delta d = \Delta m\lambda$$

$$\lambda = \frac{2\Delta d}{\Delta m}$$

4.5 INTERFEROMETERS



4.5 INTERFEROMETERS

Continuous-wave Michelson or Fourier-transform spectrograph

$$\delta = k2d = kx$$

$$I = 2I_0(1 + \cos \delta) = 2I_0(1 + \cos kx) = 2I_0 + 2I(x)$$

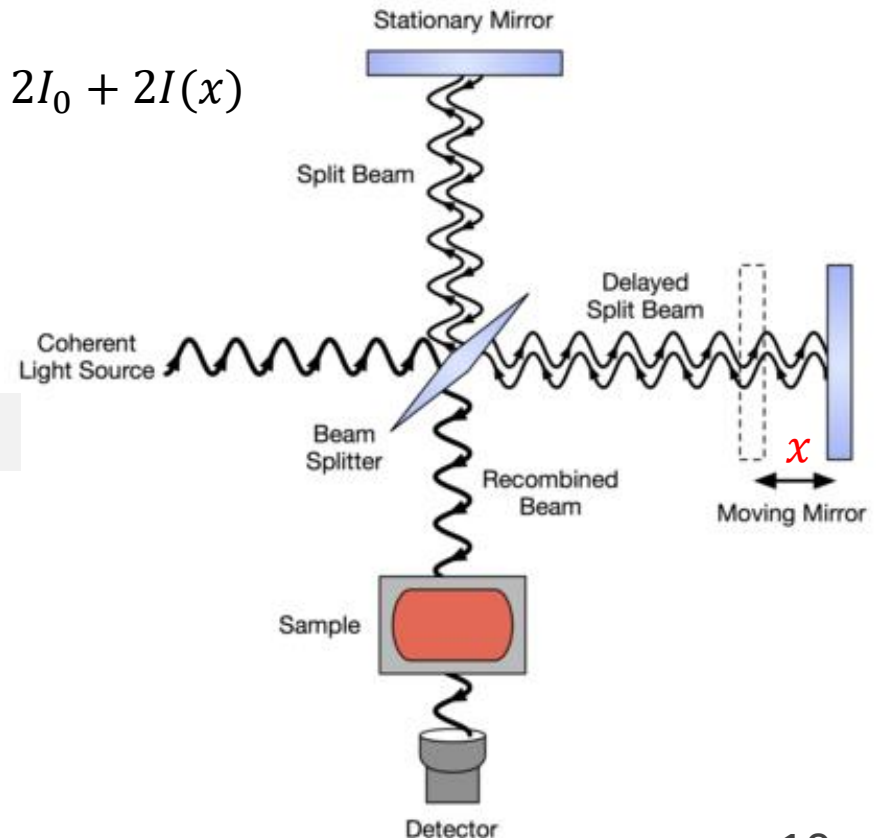
$$I(k; x) = I_0(k) \cos kx$$

$$I(\nu; x) = I_0(\nu) \cos 2\pi\nu x$$

If k is continuous in the range of $[0, \infty]$

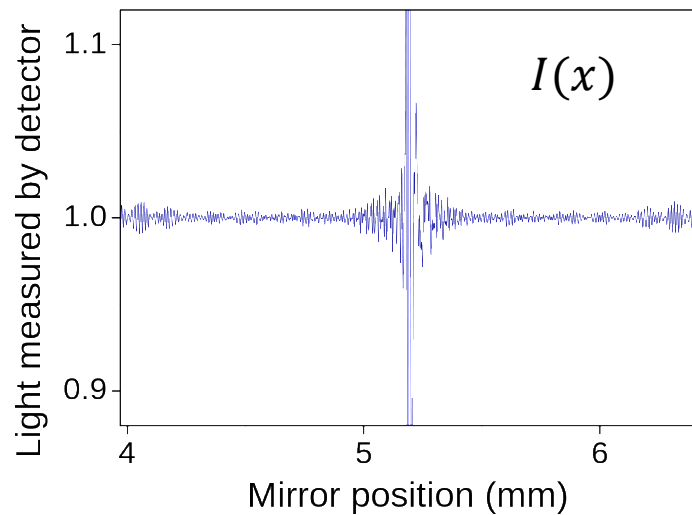
$$I(x) = \int_0^{\infty} I_0(\nu) \cos 2\pi\nu x \, d\nu$$

$$I(\nu) = \int_0^{\infty} I(x) \cos 2\pi\nu x \, dx$$

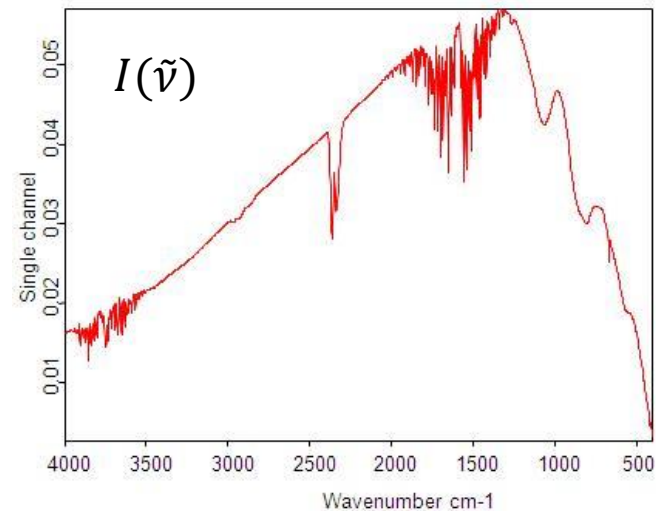


4.5 INTERFEROMETERS

Continuous-wave Michelson or Fourier-transform spectrograph



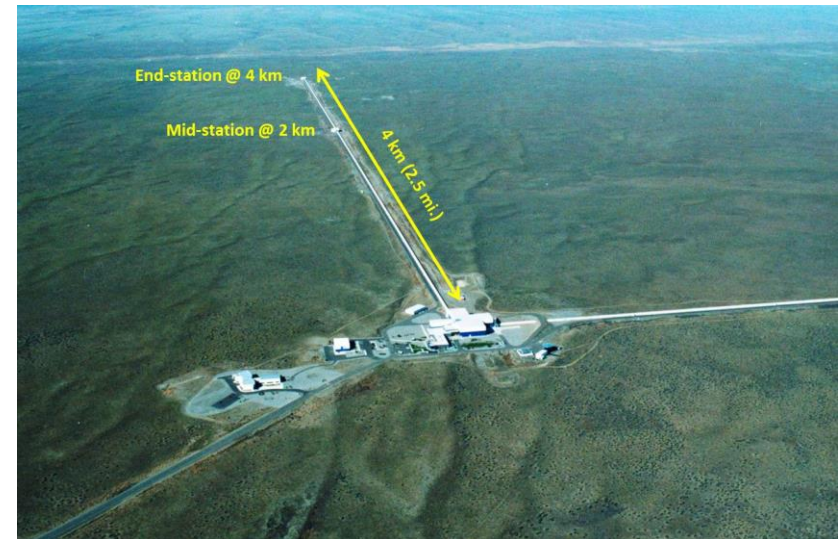
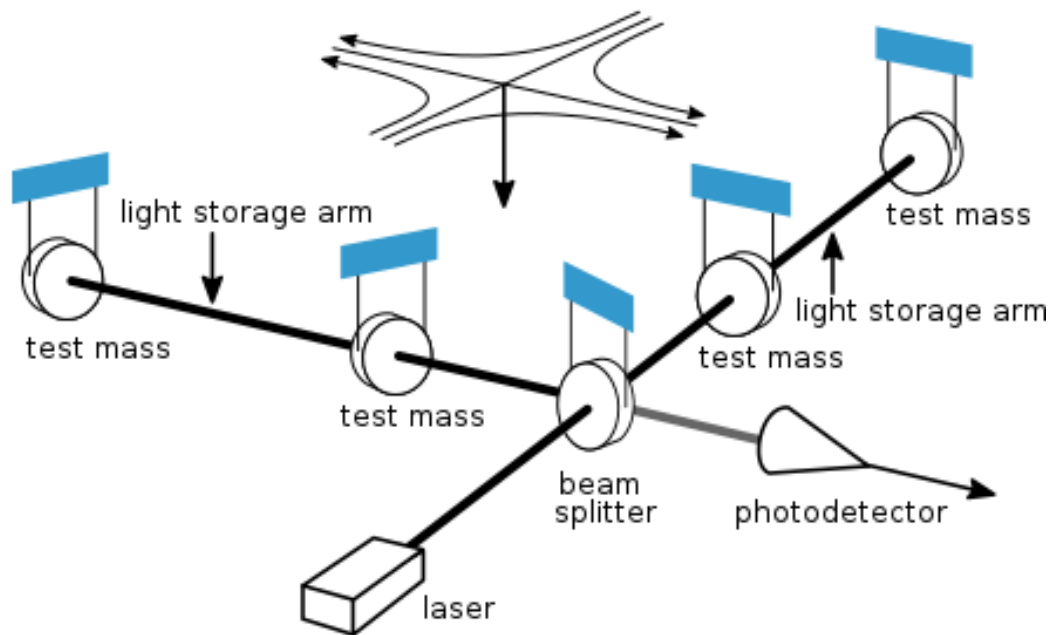
$$I(\tilde{\nu}) = \int_0^{\infty} I(x) \cos 2\pi \tilde{\nu} x \, dx$$



$$A = -\log T = -\log \frac{I_t}{I_0}$$

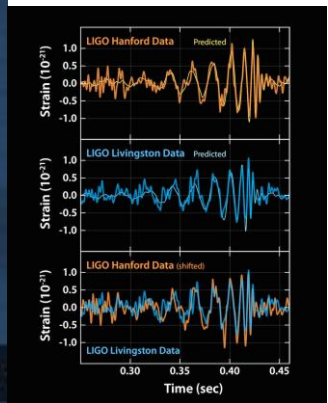
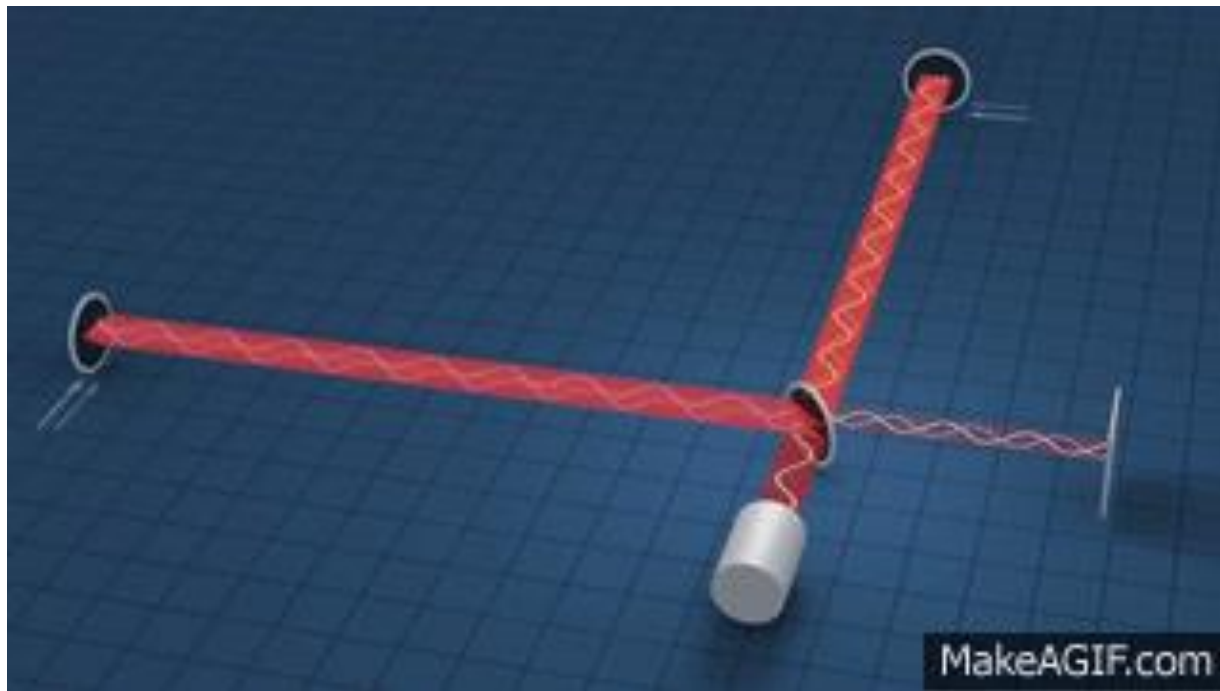
4.5 INTERFEROMETERS

gravitational-wave observatory



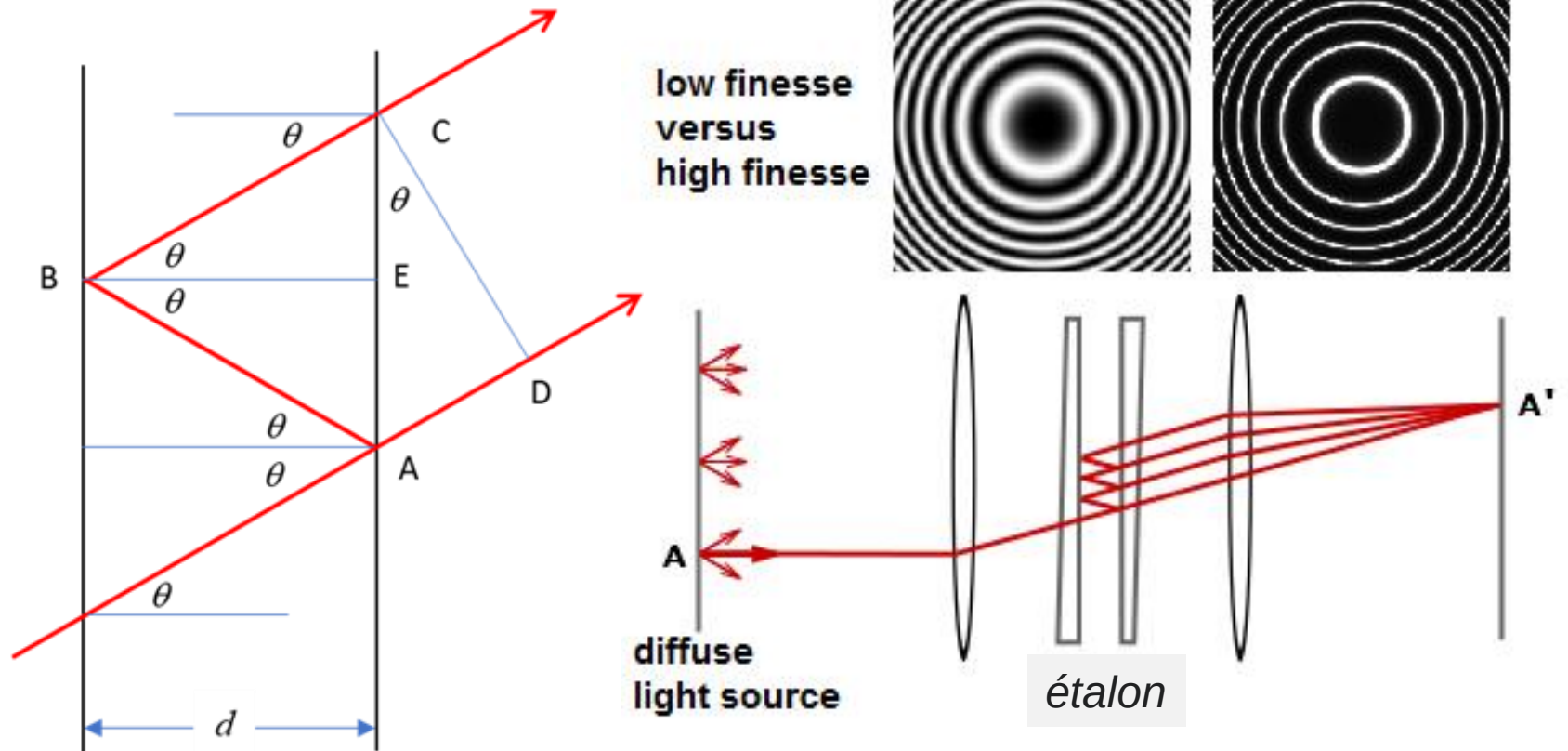
4.5 INTERFEROMETERS

gravitational-wave observatory



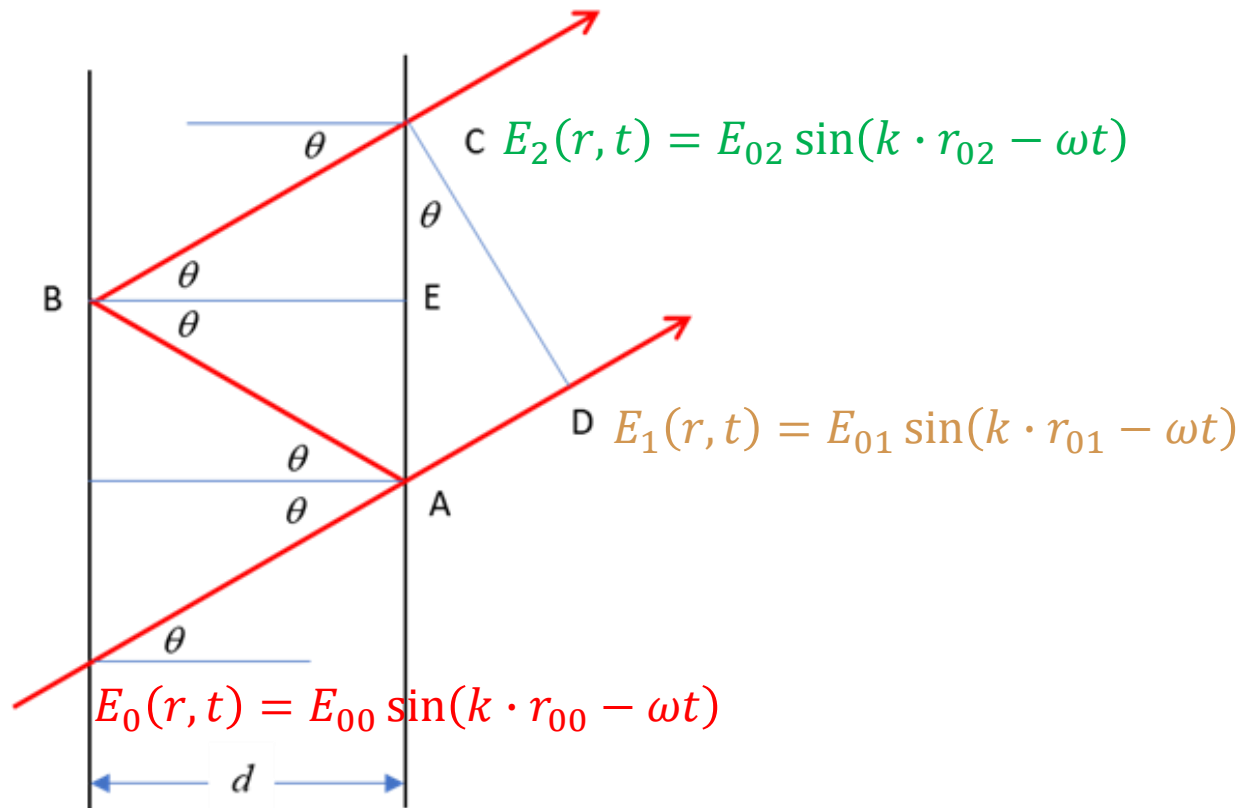
4.5 INTERFEROMETERS

Fabry-Perot interferometer



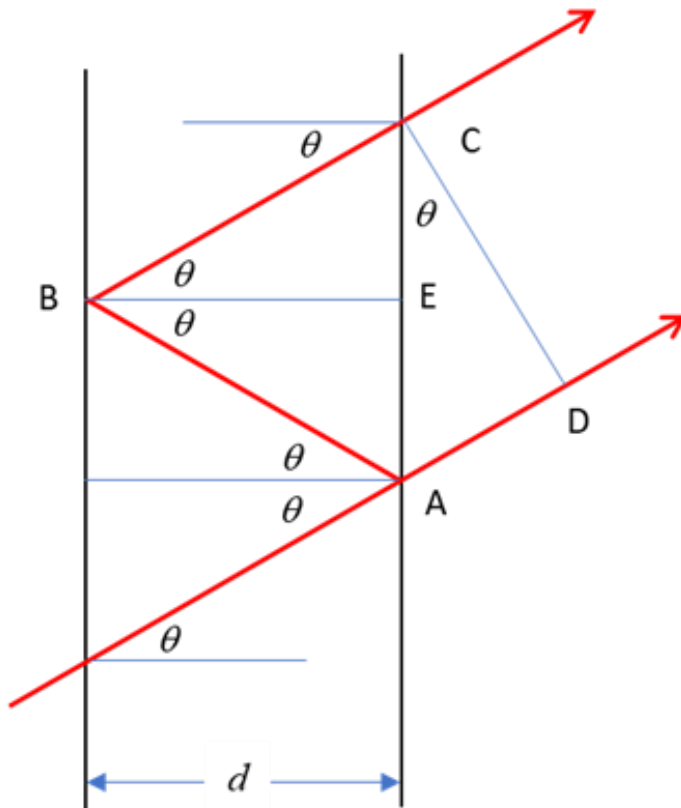
4.5 INTERFEROMETERS

Fabry-Perot interferometer



4.5 INTERFEROMETERS

Fabry-Perot interferometer



$$\Delta = AB + BC - AD$$

$$AB = BC = \frac{d}{\cos \theta}$$

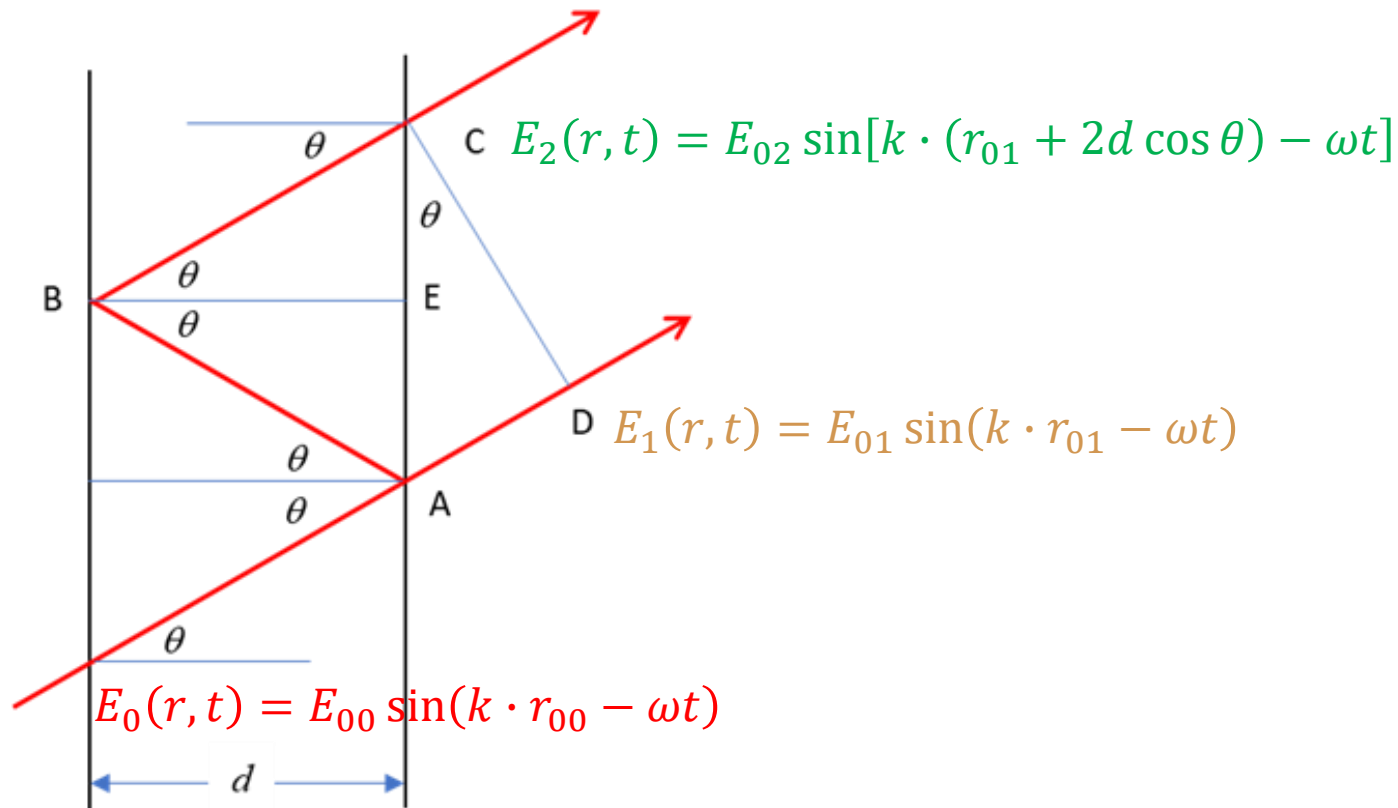
$$AD = AC \sin \theta = 2d \tan \theta \sin \theta$$

$$\Delta = \frac{2d}{\cos \theta} (1 - \sin^2 \theta) = 2d \cos \theta$$

$$\delta = k\Delta = k2d \cos \theta$$

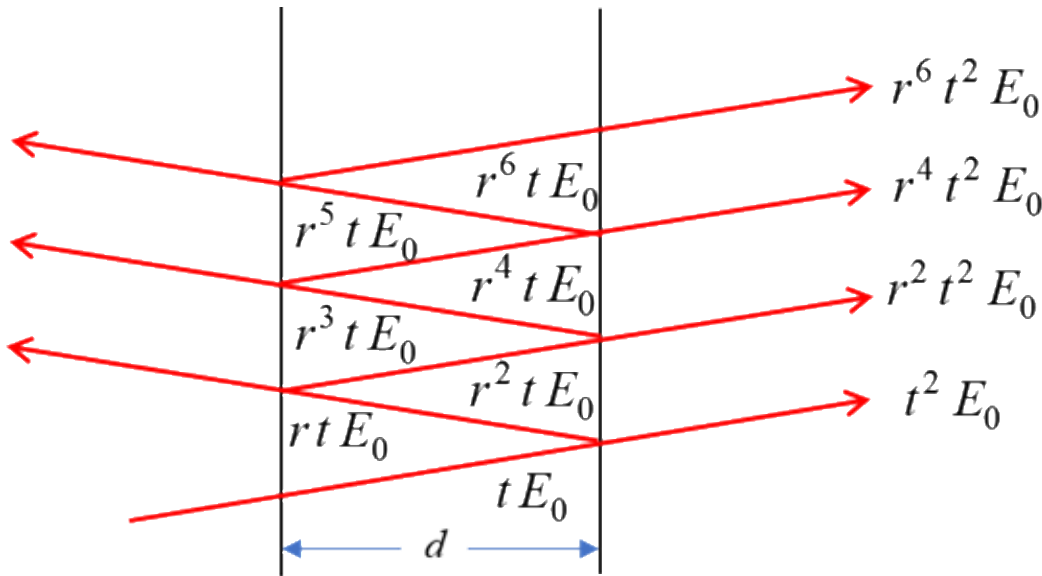
4.5 INTERFEROMETERS

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Fabry-Perot interferometer

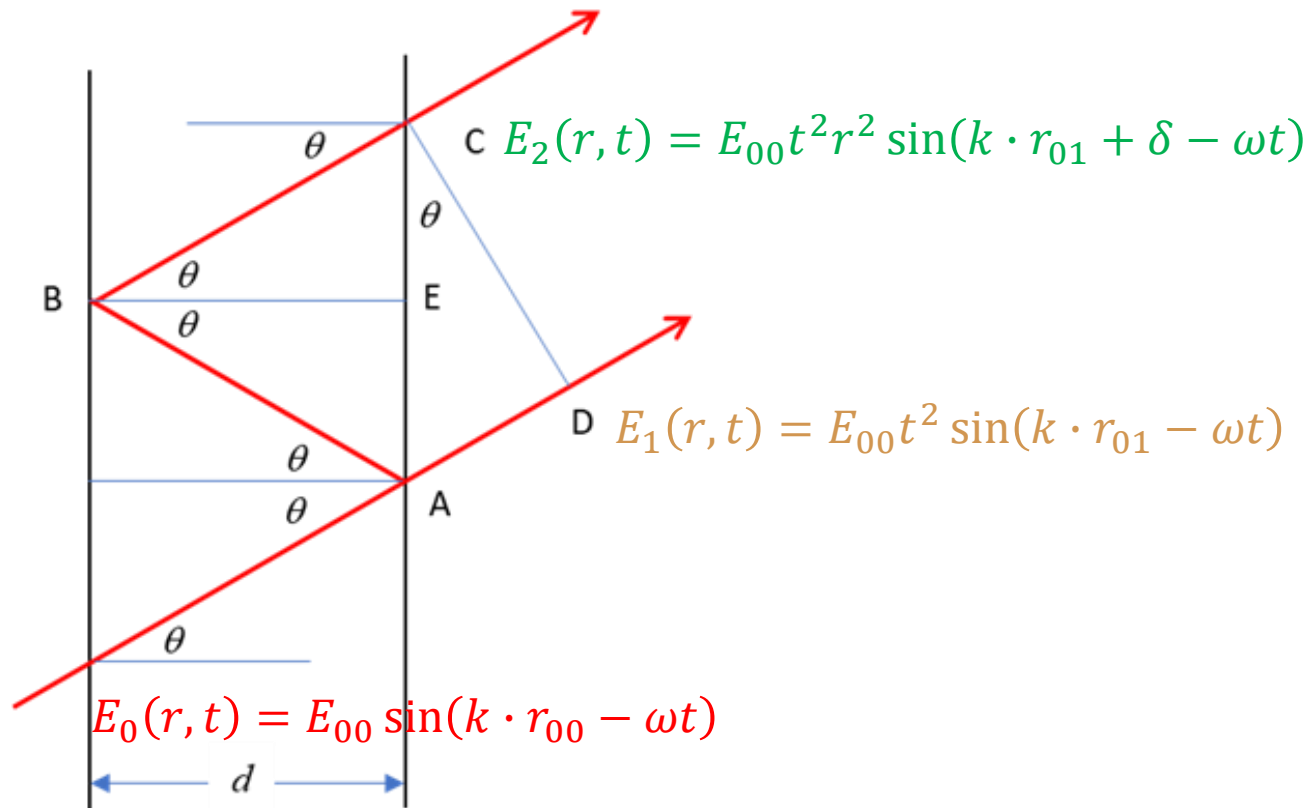


$$t = \frac{E_t}{E_0} \quad r = \frac{E_r}{E_0}$$

$$T = \frac{I_t}{I_i} = \left| \frac{E_t}{E_0} \right|^2 \quad R = \frac{I_r}{I_i} = \left| \frac{E_r}{E_0} \right|^2$$

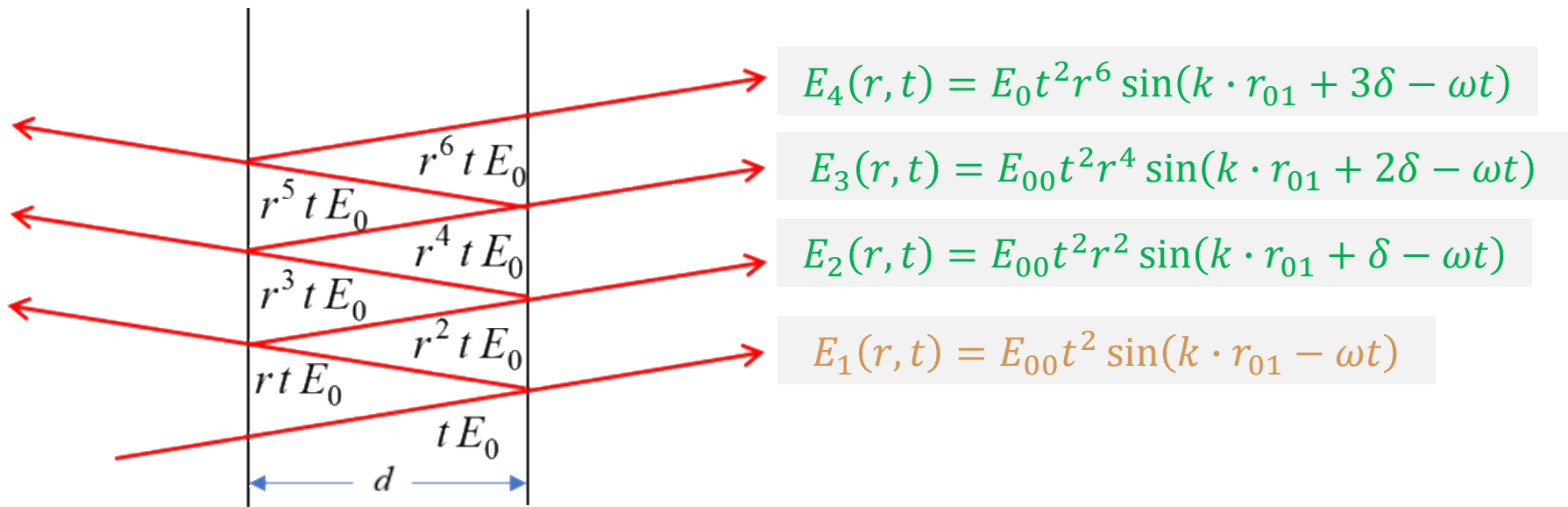
4.5 INTERFEROMETERS

Fabry-Perot interferometer



4.5 INTERFEROMETERS

Fabry-Perot interferometer



$$E_1(r, t) = E_{01} e^{ikr_1 - i\omega t}$$

$\vdots$

$$E_n(r, t) = E_{0n} e^{ikr_n - i\omega t}$$

$$E_p(r, t) = E_1 + E_2 + \cdots E_n$$

4.5 INTERFEROMETERS

Fabry-Perot interferometer

$$E_p(\mathbf{r}, t) = E_{01}e^{ikr_1 - i\omega t} + E_{02}e^{ikr_2 - i\omega t} + \dots E_{0n}e^{ikr_n - i\omega t}$$

$$E_p(\mathbf{r}, t) = E_{00}t^2e^{ikr_{01} - i\omega t} + E_{00}t^2r^2e^{ikr_{01} + i\delta - i\omega t} + \dots E_{00}t^2r^{2(n-1)}e^{ikr_{01} + i(n-1)\delta - i\omega t}$$

$$E_0 = E_{00}e^{ikr_1 - i\omega t}$$

$$E_p(\mathbf{r}, t) = t^2E_0 + t^2E_0r^2e^{i\delta} + \dots t^2E_0r^{2(n-1)}e^{i(n-1)\delta} \quad \text{Geometric progression}$$

Common ratio $r^2e^{i\delta}$

$$E_p(\mathbf{r}, t) = t^2E_0 \frac{1 - r^{2(n-1)}e^{i(n-1)\delta}}{1 - r^2e^{i\delta}} \xrightarrow{r < 1} E_p(\mathbf{r}, t) = \frac{t^2E_0}{1 - r^2e^{i\delta}}$$

4.5 INTERFEROMETERS

Fabry-Perot interferometer

$$\begin{aligned}
 I_p &= \frac{1}{2} \varepsilon_0 c |\mathbf{E}_p|^2 = \frac{1}{2} \varepsilon_0 c \mathbf{E}_p \mathbf{E}_p^* = \frac{1}{2} \varepsilon_0 c t^2 E_{00}^2 \frac{1}{1 - r^2 \cos \delta - i r^2 \sin \delta} \cdot \frac{1}{1 - r^2 \cos \delta + i r^2 \sin \delta} \\
 &= \frac{1}{2} \varepsilon_0 c t^4 E_{00}^2 \frac{1}{1 - 2r^2 \cos \delta + r^4} \\
 &= \frac{1}{2} \varepsilon_0 c t^4 E_{00}^2 \frac{1}{(1 - r^2)^2 + 4r^2 \sin^2 \frac{\delta}{2}} = \frac{1}{2} \varepsilon_0 c t^4 E_{00}^2 \frac{1}{(1 - r^2)^2 \left[1 + \frac{4r^2}{(1 - r^2)^2} \sin^2 \frac{\delta}{2} \right]}
 \end{aligned}$$

4.5 INTERFEROMETERS

Fabry-Perot interferometer

$$I_t = \frac{1}{2} \varepsilon_0 c t^4 E_{00}^2 \frac{1}{(1 - r^2)^2 \left[1 + \frac{4r^2}{(1 - r^2)^2} \sin^2 \frac{\delta}{2} \right]}$$

$$(1 - r^2)^2 = (1 - R)^2 = T^2 = t^4$$

$$\frac{1}{2} \varepsilon_0 c E_{00}^2 = I_i$$

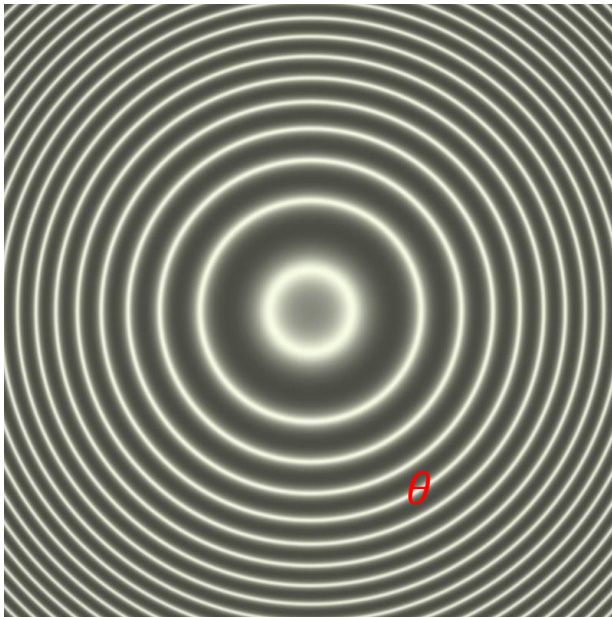
$$\frac{4r^2}{(1 - r^2)^2} = \frac{4R}{(1 - R)^2} = F$$

$$I_t = I_i \frac{1}{\left[1 + F \sin^2 \frac{\delta}{2} \right]}$$

$$T = \frac{1}{\left[1 + F \sin^2 \frac{\delta}{2} \right]}$$

4.5 INTERFEROMETERS

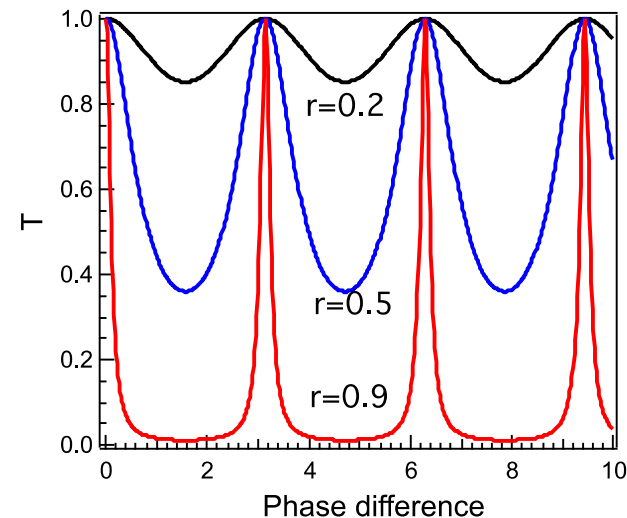
Fabry-Perot interferometer



$$\delta = k2d \cos \theta$$

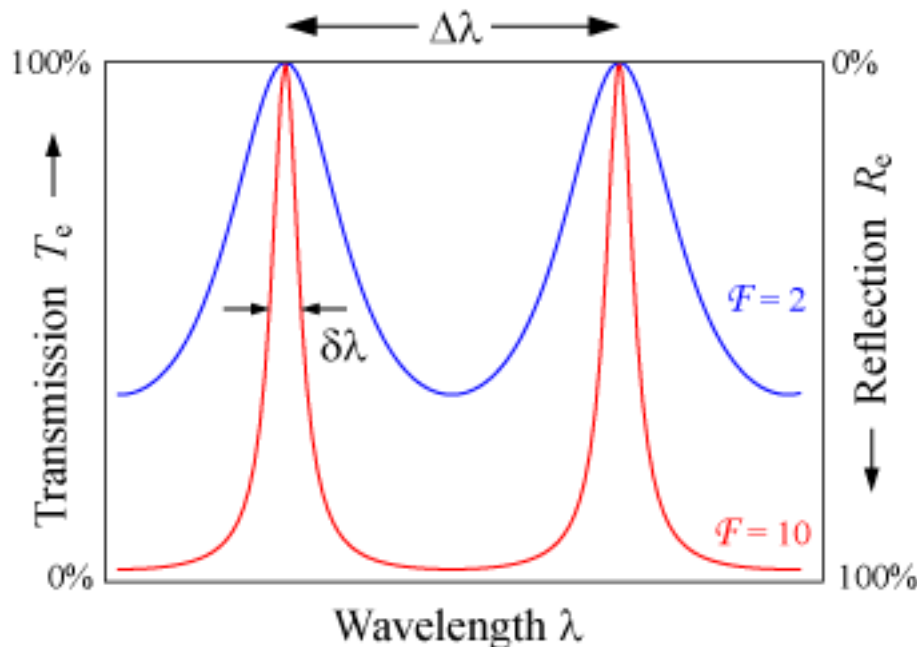
$$T = \frac{1}{\left[1 + F \sin^2 \frac{\delta}{2}\right]}$$

$$\frac{4r^2}{(1 - r^2)^2} = \frac{4R}{(1 - R)^2} = F$$



4.5 INTERFEROMETERS

Fabry-Perot interferometer



$$T = \frac{1}{\left[1 + F \sin^2 \frac{\delta}{2}\right]}$$

$$\frac{4r^2}{(1 - r^2)^2} = \frac{4R}{(1 - R)^2} = F$$

$$\delta = k2d \cos \theta = \frac{4\pi}{\lambda} d \cos \theta$$

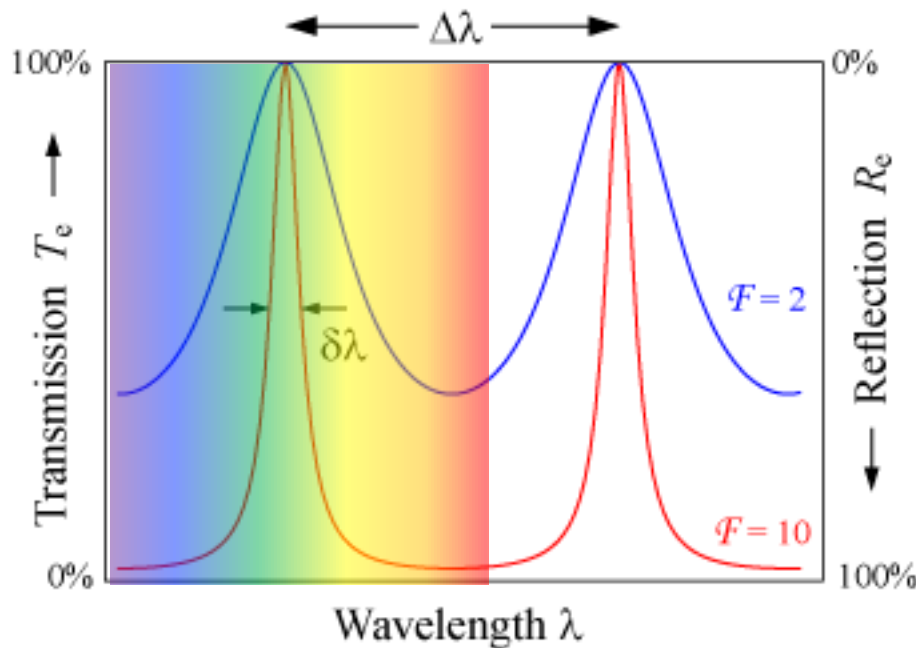
$$\text{Resolving power} \propto \sqrt{F}$$

$$\frac{2\pi}{\lambda_1} d \cos \theta = m\pi$$

$$\frac{2\pi}{\lambda_2} d \cos \theta = (m + 1)\pi$$

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Fabry-Perot interferometer



Dichroic filters